Persistence Without Contradiction: A Minimal Foundation for Existence

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Abstract

This paper establishes that persistence without globally coupled contradiction is the minimal structural requirement for determinate existence. Any entity that persists as identifiable must satisfy two independent conditions: recursive closure (stable identity under re-application of its defining boundary) and energetic viability (maintenance cost within available sustaining capacity). These are not philosophical preferences but eliminative constraints. Forms violating them cannot exist as determinate entities. We prove (1) the constraint applies to itself without regress, (2) any evaluable denial presupposes the constraint and therefore cannot succeed as a determinate denial, (3) no more primitive alternative exists, (4) the two conditions filter independently, and (5) classical paradoxes (Liar, Russell, Sorites) resolve by uniform structural diagnosis. The argument is not linear foundationalism; it is a constraint on any foundational graph. The constraint precedes physics, mathematics, and logic: it establishes what must hold for any framework to maintain stable reference.

1 Introduction: The Bedrock Claim

1.1 The Universal Presupposition

Every theoretical framework presupposes persistence without explicit justification. Physics tracks systems across time. Mathematics maintains reference across inference. Logic requires stable propositions. Computation demands persistent states. Language assumes continuity of reference. Measurement presumes observables do not dissolve mid-act.

Yet persistence itself is rarely analyzed as a first-class constraint. It functions as an unexamined background condition.

1.2 The Prior Question

This paper investigates: *What structural conditions must hold for anything to persist as re-identifiable?*

The answer is eliminative. Forms that violate these conditions do not persist, because violation structurally prevents persistence.

Anti-Circularity Note: Persistence is not identical to identity. Identity describes the relation of a thing to itself ($A = A$); persistence is the structural constraint that makes any identity claim evaluable over transformation or time. Without persistence, the terms of an identity relation dissolve before the relation can obtain. Thus, persistence is the precondition for identity, not a synonym for it.

1.3 Scope Clarification and Ontological Bridge

This paper establishes a structural constraint that functions as an eliminative ontology.
Definition 1 (Existence as Determinate Re-identifiability)**.** *A candidate form $\psi \in S$* exists *if and only if:*

1. *There exists a selective criterion $C_\psi : S \to \{0, 1\}$ such that $C_\psi(\psi) = 1$ and $\exists \phi \in S$ with $C_\psi(\phi) = 0$,*
2. *There exists a nonempty admissible transformation family $\Delta(\psi)$ (Definition 7) such that $\forall \delta \in \Delta(\psi)$, $C_\psi(\delta(\psi)) = 1$,*
3. *The maintenance cost functional W_ψ satisfies energetic viability (Definition 12),*
4. *C_ψ is stable under the role-constitutive transformations in $\Delta(\psi)$ and is not defined by explicit reference to the evaluation context or this paper's own definitions.*

If any condition fails, the term "ψ exists" lacks determinate referent.

This is not a definitional reduction of existence to cognition or language. The term "re-identifiability" denotes a structural property: the existence of selective invariants stable under the role-constitutive transformations of a candidate form. The claim is eliminative: if no such invariant structure exists, there is no determinate object for any ontology (physical, mathematical, or semantic) to be about.

This paper does not claim this definition exhausts all uses of the term "exists." It proposes a minimal eliminative criterion for determinate existence: whatever can be stably referred to as a form across admissible transformations. The constraint therefore does not merely restrict what we can track. It restricts what can count as a determinate existent at all. The paper remains eliminative: it does not enumerate particular beings, it rules out structural non-being.

1.4 Claimability, Witness Conditions, and the Use–Mention Gap

A common evasion concedes that discourse requires determinacy, but insists reality might contain "existents" that are not determinately re-identifiable. This is often expressed as a

use–mention split:

> The existential *sentence* is determinate, but the alleged *referent* need not be.

This section closes that escape. The closure is not idealism, and it is not a reduction of being to language. It is a structural constraint on truth-evaluable existential claims: if an existential claim is genuinely about something, it must support nontrivial witness conditions stable under the role-constitutive transformations required to apply the predicate. Otherwise the claim has no determinate truth conditions and is not an ontological claim.

Definition 2 (Truth-Evaluable Existential Commitment)**.** *An existential claim of the form $\exists x\, P(x)$ is* truth-evaluable as an ontological claim *only if there exists a nontrivial witness condition W_P such that:*

1. *(**Nontriviality**) W_P distinguishes at least one candidate witness from at least one non-witness,*
2. *(**Stability**) W_P remains stable under the role-constitutive transformations required to apply or evaluate P.*

If no such witness condition exists, then "$\exists x\, P(x)$" is not false but non-determinate as an ontological assertion.

Lemma 3 (Aboutness Requires Witnessable Selection)**.** *If an existential claim "$\exists x\, P(x)$" is truth-evaluable as* about *some class of entities, then it presupposes a selective invariant sufficient to distinguish witnesses from non-witnesses under the role-constitutive transformations of P.*

Proof. To be truth-evaluable, "$\exists x\, P(x)$" must have determinate truth conditions. In standard semantics, it is true only if there exists at least one witness a such that $P(a)$ holds, and false only if no such witness exists. If there is no nontrivial and stable way to distinguish a witness from a non-witness under the operations constitutive of applying or evaluating P, then the distinction between "there exists a witness" and "there exists none" collapses into indeterminacy. Therefore truth-evaluable existential aboutness presupposes witnessable selection stable under the role-constitutive transformations of P. □ □

Theorem 4 (Closure of the Use–Mention Gap for Existential Claims)**.** *Let C be an existential claim of the form $\exists x\, P(x)$. Any attempt to maintain that C is a determinate ontological claim while asserting that all alleged witnesses are not determinately re-identifiable yields one of two outcomes:*

1. *C is truth-evaluable only because P admits a nontrivial stable witness condition, in which case the alleged witnesses are re-identifiable at least to the degree required for determinate existence (Definition 1),*
2. *or C lacks determinate truth conditions and is not an ontological claim (Definition 2).*

Thus the "claim vs. referent" distinction cannot be used to evade the re-identifiability constraint while retaining an evaluable existential assertion.

Proof. Assume C is a determinate ontological claim. Then by Lemma 3, C presupposes a nontrivial stable witness condition for P under the role-constitutive transformations required to apply or evaluate P. But a stable witness condition induces a selective criterion that distinguishes at least one witness from a non-witness and remains stable under admissible transformations. This is precisely the minimal structure demanded by determinate re-identifiability in Definition 1.

If no such witness condition exists, then C is not truth-evaluable as an ontological claim by Definition 2. Either way, the attempted use–mention escape fails. □ □

Remark 5 (Vagueness and Ineffability Do Not Defeat Witness Conditions). Vagueness typically denotes a fuzzy boundary, not the absence of any boundary. Fuzzy membership still induces nontrivial selection (e.g., via thresholds, supervaluation families, or index parameters). Ineffability denotes limits of description, not failure of individuation. An experience may be indescribable in full content while remaining identifiable as an event-token under role-constitutive operations (occurrence, causal trace, report-triggering, subject-binding). Therefore vagueness and ineffability do not supply a counterexample unless redefined to mean "not even witnessable," which collapses the existential claim into non-determinacy.

2 Structural Foundations

We fix minimal primitives sufficient to state and test the constraint without importing a specific domain ontology.

2.1 Minimal Primitives and Admissibility

Definition 6 (State Space and Forms)**.** *A* state space *is a nonempty set S of candidates. A* form *is any $\psi \in S$ together with an* identity condition $\equiv_\psi$ *that specifies which states count as the same form across admissible transformations.*

Definition 7 (Admissible Transformations)**.** *For each form $\psi \in S$, the admissible transformation family $\Delta(\psi)$ is a nonempty set of endomaps $\delta : S \to S$ fixed by $\equiv_\psi$ and by the publicly checkable role-constitutive operations required for ψ to count as an instance of its claimed structural role (Proposition 8), such that:*

1. *(**Identity preservation**) $\forall \delta \in \Delta(\psi)$, $\delta(\psi) \equiv_\psi \psi$,*
2. *(**Closure**) if $\delta_1, \delta_2 \in \Delta(\psi)$ then $\delta_2 \circ \delta_1 \in \Delta(\psi)$,*
3. *(**Non-gerrymandering**) $\Delta(\psi)$ is fixed prior to evaluating persistence claims about ψ and may not be chosen post hoc to preserve ψ against counterexamples.*

Proposition 8 (Role Determines Admissible Transformations)**.** *If a form ψ claims to occupy structural role R (e.g., truth-bearer, set, number, physical object), then $\Delta(\psi)$ must include all transformations constitutive of role R. A form cannot stipulate away role-constitutive transformations while claiming to occupy that role. This is not an external addition to $\equiv_\psi$; it is a constraint on any $\equiv_\psi$ that purports to define an instance of role R. For example:*

- *If ψ is a truth-bearer, $\Delta(\psi)$ must include evaluation operations.*

- *If ψ is a set, $\Delta(\psi)$ must include membership tests.*
- *If ψ is a computation, $\Delta(\psi)$ must include step transitions.*

Proof. Suppose form ψ claims to occupy structural role R but stipulates that $\Delta(\psi)$ excludes transformations δ_R that are constitutive of role R.

Then ψ's identity conditions require:

1. ψ is-an-R (the role claim)
2. Being-an-R requires $\delta_R \in \Delta(\psi)$ (role-constitutive transformations)
3. $\delta_R \notin \Delta(\psi)$ (the stipulation)

From (1) and (2): $\delta_R \in \Delta(\psi)$. From (3): $\delta_R \notin \Delta(\psi)$.

Therefore ψ requires both $\delta_R \in \Delta(\psi)$ and $\delta_R \notin \Delta(\psi)$. Let $P(\psi) :=$ "$\delta_R \in \Delta(\psi)$". Then ψ requires both $P(\psi)$ and $\neg P(\psi)$.

By Definition 16 (Section 4), this is a globally coupled contradiction internal to the identity conditions of ψ. In Section 4 we prove that globally coupled contradiction eliminates persistence (Theorem 25). Therefore, a form cannot stipulate away role-constitutive transformations while claiming to occupy that role. □ □

Definition 9 (Admissible Distinction Operators)**.** *A distinction operator $\hat{\delta}$ is* admissible *for ψ if it is not a vacuous identity map and if its boundary is invariant under admissible transformations:*

1. *(**Nontriviality**) there exists $\phi \in S$ such that $\hat{\delta}(\phi) \neq \phi$,*
2. *(**Commutation on** ψ) $\forall \delta \in \Delta(\psi)$, $\hat{\delta}(\delta(\psi)) = \delta(\hat{\delta}(\psi))$,*
3. *(**Stability**) $\hat{\delta}$ induces a determinate boundary on ψ under iteration within $\Delta(\psi)$.*

We take $\hat{\delta} : S \to S$ to be an endomap.

2.2 The Persistence Filter

Definition 10 (Persistence Filter)**.** *A form ψ* persists as the same form *(relative to $\equiv_\psi$) only if:*

$$\forall n \in \mathbb{N},\ \forall \delta \in \Delta(\psi) :\ \psi \equiv_\psi (\hat{\delta} \circ \delta)^n(\psi)\ \wedge\ \Phi_\psi(n) \geq W_\psi(\delta, n), \tag{1}$$

where $\equiv_\psi$ is the identity condition of ψ (Definition 6), $\hat{\delta}$ is admissible for ψ (Definition 9), $\Phi_\psi : \mathbb{N} \to \mathbb{R}_{\geq 0}$ is the sustaining capacity available to maintain ψ's boundary across n updates, and $W_\psi : \Delta(\psi) \times \mathbb{N} \to \mathbb{R}_{\geq 0}$ is the maintenance cost required to preserve that boundary under admissible iteration. Persistence is invariance under admissible iteration, not single-step resemblance.

The first condition captures **recursive closure**: the form remains self-identical under re-application of its defining distinction. The second captures **energetic viability**: the form maintains its boundary without exceeding available resources.

3 Energetic Viability and Solvency

Remark 11 (Substrate-Neutral Cost). The energetic viability condition is not a smuggling of physicalism, but a recognition of information-theoretic and structural limits. "Cost" here refers to representational and proof-theoretic resources: finite describability, bounded verification, and bounded compression.

- **Intensional forms (rules):** A recursive rule like the Peano Axioms has finite descriptive complexity. The cost to maintain the *rule* is constant. Thus, mathematical rules persist.
- **Extensional forms (completed totalities):** A form defined as "the completed, actualized list of all natural numbers" requires unbounded representational maintenance. To maintain determinate reference to such an object in any finite medium requires unbounded capacity.

This paper is neutral on platonism; it shows that any form requiring unbounded representational maintenance cannot persist as a determinate referent in any finite medium. Therefore: mathematical *rules* persist; completed extensional infinities do not exist as determinate forms in finite substrates.

Definition 12 (Maintenance Cost Functional)**.** *Fix ψ and its admissible family $\Delta(\psi)$. A maintenance cost functional is any map*

$$W_\psi : \Delta(\psi) \times \mathbb{N} \to \mathbb{R}_{\geq 0}$$

such that $W_\psi(\delta, n)$ is the cost required to preserve ψ's boundary through n successive admissible updates under δ. A sustaining capacity functional is any map

$$\Phi_\psi : \mathbb{N} \to \mathbb{R}_{\geq 0}$$

representing available maintenance capacity across n updates. Energetic viability means:

$$\forall n \in \mathbb{N},\ \forall \delta \in \Delta_R(\psi) :\ \Phi_\psi(n) \geq W_\psi(\delta, n),$$

where $\Delta_R(\psi)$ are the role-constitutive transformations.

Lemma 13 (Finite Substrate Bound)**.** *For any physically realizable or finitely representable system, there exists a finite upper bound $\Phi_{\max}(n) < \infty$ on maintainable protected capacity across n updates. Consequently, any form whose maintenance cost satisfies $\limsup_{n\to\infty} W_\psi(\delta, n) = \infty$ fails energetic viability relative to any finite substrate.*

Proof. If a system is physically realizable or finitely representable, then its realizations correspond to descriptions over a finite alphabet with bounded length (or bounded state cardinality) at each stage of representation. Hence the number of distinct encodable states is finite at each n, implying a finite upper bound on storable protected information and therefore on sustaining capacity across n updates. Equivalently, any finite-state medium admits only finitely many distinguishable configurations, so capacity cannot diverge within that medium. Therefore $\Phi_{\max}(n) < \infty$ for each n and unbounded maintenance cost cannot be met by a finite substrate. □ □

4 The Impossibility of Persisting Contradiction

Definition 14 (Re-identification)**.** *A form $\psi \in S$ is* re-identifiable under $\Delta(\psi)$ *if and only if there exists a criterion function $C : S \to \{0, 1\}$ such that:*

1. *(**Application**) $C(\psi) = 1$,*
2. *(**Selectivity**) $\exists \phi \in S$ with $\phi \neq \psi$ such that $C(\phi) = 0$,*
3. *(**Stability**) $\forall \delta \in \Delta(\psi)$: $C(\delta(\psi)) = 1$.*

If no such criterion exists, ψ is not re-identifiable and therefore does not persist as a determinate form.

Theorem 15 (Evaluation Requires Stable Targeting)**.** *Any evaluable proposition presupposes re-identification of its target across the evaluation map. Therefore, "persistence" is not inferred from identity; it is the precondition that makes identity claims evaluable at all.*

Proof. Let P be a proposition about ψ. To evaluate P, the referent ψ must remain targeted from the moment of utterance/inscription to the moment of truth-value assignment. If ψ transforms into non-ψ (relative to the selection criterion of P) during this interval, P becomes indeterminate or referentially void. Thus, stable targeting (re-identification) is a structural prerequisite for evaluation. □ □

Definition 16 (Globally Coupled Contradiction (GCC))**.** *A form ψ contains a globally coupled contradiction if there exist predicates P such that ψ's identity conditions require both $P(\psi)$ and $\neg P(\psi)$ to hold, and this requirement is inseparable in the following sense: there exists no witness structure π such that:*

1. ***Localization:** π assigns P and $\neg P$ to disjoint structural or spatial regions of ψ while preserving ψ's identity under $\Delta(\psi)$, or*
2. ***Indexing:** π assigns P and $\neg P$ to distinct indices (temporal moments, modal contexts, or parameter values) while preserving ψ's identity under $\Delta(\psi)$, or*
3. ***Staging:** π assigns P and $\neg P$ to distinct stages of a process while preserving ψ's identity under $\Delta(\psi)$, or*
4. ***Confinement:** π isolates the contradiction to a proper subdomain while maintaining a consistent superstructure that grounds ψ's persistence under $\Delta(\psi)$.*

If no such witness π exists that is stable under $\Delta(\psi)$, the contradiction is globally coupled. By 'identity conditions' we mean the minimal structure required to re-identify ψ as the same form under $\Delta(\psi)$, not merely propositions about ψ in an external metalanguage.

Stability means: for all $\delta \in \Delta(\psi)$, $\pi(\delta(\psi)) = \pi(\psi)$ up to the equivalence relation induced by $\equiv_\psi$.

Operationally: a contradiction is globally coupled for ψ if no stable partition witness π can be exhibited that prevents P and $\neg P$ from coupling at the identity-grounding layer while preserving ψ under $\Delta(\psi)$. In this paper, GCC is established by showing failure of all standard partition modes (Remark 18) relative to $\Delta(\psi)$. Any proposed nonstandard separation must

be provided as an explicit witness π stable under $\Delta(\psi)$; otherwise it is not a determinate alternative.

Definition 17 (Partition Witness)**.** *A contradiction is partitioned if it admits a witness structure π satisfying one of the conditions (Localization, Indexing, Staging, or Confinement) in Definition 16.*

Remark 18 (Standard Partition Modes and Burden of Production)*.* In practice, attempts to prevent contradiction from coupling at the identity-grounding layer take one of four standard forms: localization (structural separation), indexing (parameter or context separation), staging (process-step separation), or confinement (hierarchy separation). This paper does not require the claim that these four are exhaustive. Instead, the operational standard is:

GCC is established when all standard partition modes fail *under the role-constitutive transformations* $\Delta(\psi)$. If an objector claims a further separation mode exists, the burden is to *exhibit* a concrete witness structure π that: (i) separates P from $\neg P$, and (ii) is stable under $\Delta(\psi)$ as required by Proposition 19. Absent such an exhibited witness, the claim "there exists some other separation" is non-operational.

Proposition 19 (Stability Requirement for Partition Witness)**.** *If a contradiction in ψ is claimed to be partitioned, then there exists a partition witness π (Definition 17) that is stable under $\Delta(\psi)$.*

Proof. A partition witness π is introduced to prevent the contradictory requirements $P(\psi)$ and $\neg P(\psi)$ from coupling at the identity-grounding level of ψ while preserving ψ under admissible transformations. If π is not stable under $\Delta(\psi)$, then there exists $\delta \in \Delta(\psi)$ such that π fails to preserve its separation role for $\delta(\psi)$, and the contradiction re-couples under admissible iteration. Hence any genuine partition must be stable under $\Delta(\psi)$. □ □

4.1 The Eliminative Proofs

Lemma 20 (Criterion Implies Partition)**.** *Let S be a nonempty state space and let $C : S \to \{0, 1\}$ be a criterion. If C is selective for $\psi \in S$ in the sense that $C(\psi) = 1$ and there exists $\phi \in S$ with $\phi \neq \psi$ and $C(\phi) = 0$, then C induces a nontrivial partition of S:*

$$S_1 := C^{-1}(1), \qquad S_0 := C^{-1}(0),$$

with $\psi \in S_1$ and $S_0 \neq \varnothing$. Conversely, any nontrivial partition (S_1, S_0) with $\psi \in S_1$ and $S_0 \neq \varnothing$ defines a selective criterion by $C(x) = 1$ iff $x \in S_1$.

Proof. If $C(\psi) = 1$, then $\psi \in C^{-1}(1) = S_1$. If there exists $\phi \neq \psi$ with $C(\phi) = 0$, then $\phi \in C^{-1}(0) = S_0$, so $S_0 \neq \varnothing$. Since codomain is $\{0, 1\}$, $S_1 \cap S_0 = \varnothing$ and $S_1 \cup S_0 = S$, hence (S_1, S_0) is a partition and nontrivial because $S_0 \neq \varnothing$.

Conversely, given a partition (S_1, S_0) with $\psi \in S_1$ and $S_0 \neq \varnothing$, define $C(x) = 1$ iff $x \in S_1$. Then $C(\psi) = 1$ and for any $\phi \in S_0$, $C(\phi) = 0$ with $\phi \neq \psi$ since S_0 and S_1 are disjoint. □ □

Theorem 21 (Re-identification Implies an Identity-Grounding Boundary Witness)**.** *If ψ is re-identifiable under $\Delta(\psi)$, then there exists a boundary witness π_ψ at the identity-grounding layer such that:*

1. *π_ψ induces a nontrivial separation between "ψ-states" and "non-ψ-states",*
2. *π_ψ is stable under $\Delta(\psi)$ in the sense that for all $\delta \in \Delta(\psi)$, the witness continues to separate $\delta(\psi)$ from at least one alternative,*
3. *π_ψ is sufficient to ground a selective invariant criterion for ψ under admissible iteration.*

Proof. Assume ψ is re-identifiable under $\Delta(\psi)$. Then there exists a criterion $C : S \to \{0, 1\}$ satisfying Definition 14. By Lemma 20, C induces a nontrivial partition (S_1, S_0) with $\psi \in S_1$ and $S_0 \neq \varnothing$.

Because C is stable under $\Delta(\psi)$, $\delta(\psi) \in S_1$ for all $\delta \in \Delta(\psi)$. Thus the boundary between S_1 and S_0 is invariant on ψ under admissible transformations.

Define π_ψ as the induced identity-grounding boundary specification:

$$\pi_\psi := \langle S_1, S_0, \Delta(\psi)\rangle,$$

interpreted as the minimal witness structure that fixes what counts as "inside" (S_1) versus "outside" (S_0) for the purpose of re-identifying ψ across admissible iteration.

Nontriviality holds since $S_0 \neq \varnothing$. Stability holds since admissible evolution preserves membership of ψ-iterates in S_1. Therefore π_ψ is a boundary witness sufficient to ground a selective invariant criterion for ψ under $\Delta(\psi)$. □ □

Theorem 22 (Determinacy Forces Selective Invariance)**.** *If ψ exists as a determinate form in the sense of Definition 1, then there exists a nontrivial criterion $C : S \to \{0, 1\}$ that distinguishes ψ from at least one alternative and remains stable under the role-constitutive transformations $\Delta(\psi)$.*

Proof. By Definition 1, existence of ψ entails: (i) there exists a selective criterion C_ψ with $C_\psi(\psi) = 1$ and $\exists\phi$ with $C_\psi(\phi) = 0$, and (ii) stability of that criterion under the admissible family $\Delta(\psi)$, in particular under the role-constitutive transformations. Thus C_ψ is nontrivial, distinguishing, and stable under $\Delta(\psi)$. □ □

Lemma 23 (GCC Implies No Stable Re-identification Criterion)**.** *If ψ contains a globally coupled contradiction (Definition 16), then there exists no re-identification criterion $C : S \to \{0, 1\}$ satisfying Definition 14.*

Proof. Assume for contradiction that such a criterion C exists. Then C is selective and stable under $\Delta(\psi)$. By Lemma 20, selectivity of C induces a nontrivial partition (S_1, S_0) with $\psi \in S_1$ and $S_0 \neq \varnothing$. By stability, this partition is preserved under admissible transformations in the sense that $\forall\delta \in \Delta(\psi)$, $\delta(\psi) \in S_1$.

By Theorem 21, re-identifiability of ψ under $\Delta(\psi)$ entails the existence of a stable boundary witness π_ψ at the identity-grounding layer sufficient to support selective invariance. Such a boundary witness functions as a partition witness in the sense of Definition 17: it separates admissible ψ-states from at least one alternative in a manner stable under $\Delta(\psi)$.

But GCC (Definition 16) asserts that no such stable witness exists that can separate contradictory grounding requirements while preserving ψ under $\Delta(\psi)$. Contradiction.

Therefore no such re-identification criterion exists. □ □

Lemma 24 (Contradiction Eliminates Re-identifiability)**.** *Let ψ be a form with admissible transformations $\Delta(\psi)$. If ψ contains a globally coupled contradiction, then ψ is not re-identifiable under $\Delta(\psi)$.*

Proof. By Lemma 23, globally coupled contradiction implies that no stable selective criterion $C : S \to \{0, 1\}$ exists satisfying Definition 14. Therefore ψ is not re-identifiable under $\Delta(\psi)$. □ □

Theorem 25 (Persistence Requires Non-Contradiction)**.** *Let ψ be a form with admissible transformations $\Delta(\psi)$. If ψ contains a globally coupled contradiction, then ψ cannot persist as an identifiable form.*

Proof. By Lemma 24, a globally coupled contradiction eliminates the structural conditions required for re-identification of ψ under $\Delta(\psi)$.

By the Persistence Filter (Definition 10), persistence as an identifiable form requires recursive closure of the boundary distinction and, minimally, structural re-identifiability (Definition 14), since the persistence condition $\psi \equiv_\psi (\hat{\delta} \circ \delta)^n(\psi)$ is meaningful only when ψ remains the same form across admissible iteration.

A globally coupled contradiction at the identity-grounding layer collapses the selection boundary required for nontrivial reference, yielding either triviality (everything counts) or indeterminacy (nothing counts); neither supports determinate persistence.

Therefore, if re-identification fails, persistence fails. Hence globally coupled contradiction prevents persistence. □ □

Corollary 26 (Failure of Re-identifiability Implies Non-Existence)**.** *If ψ is not re-identifiable under $\Delta(\psi)$, then ψ does not exist as a determinate form in the sense of Definition 1.*

Proof. By Definition 1, existence as a determinate form requires re-identifiability via a selective, stable criterion. If re-identifiability fails, the existence conditions fail. □ □

5 Independence of Constraints

A critical structural question arises: Is energetic viability merely a corollary of recursive closure, or does it perform independent eliminative work? We prove independence by demonstrating a form that satisfies recursive closure (logically consistent) yet fails energetic viability (resource insolvent).

Theorem 27 (Independence of Energetic Viability)**.** *There exists a form ψ such that recursive closure holds for ψ (it is consistent and boundary-stable as a rule) but energetic viability fails for ψ (it is insolvent relative to any finite sustaining capacity).*

Proof. Define a form ψ (the "Perfect Sentinel") by the identity condition:

$$\text{Identity}(\psi): \text{ "Maintain a complete, lossless record of all my prior states forever."}$$

(1) Recursive closure holds. The identity condition is internally coherent: it does not require $P(\psi)$ and $\neg P(\psi)$ simultaneously at the grounding level. Under re-application of its defining boundary distinction, it returns the same rule-level criterion: "keep a complete lossless history." Thus the rule-level form is stable under iteration.

(2) Energetic viability fails. Let the state at step t be S_t and define the update $S_{t+1} = S_t \oplus h(S_t)$ where $\oplus$ concatenates and h is a lossless encoding. Then record length is strictly increasing whenever the system changes:

$$\|S_{t+1}\| > \|S_t\|.$$

By Lemma 13, any finite substrate has bounded sustaining capacity across indefinite iteration. But the maintenance cost $W_\psi(\delta, t)$ required to preserve a growing lossless archive is monotonically increasing in t (at minimum, storage and protection costs increase with record size). Therefore there exists T such that for all $t > T$, $W_\psi(\delta, t) > \Phi_{\max}(t)$. At that point, ψ must either discard information (violating its identity condition) or collapse as that form.

Hence ψ satisfies recursive closure but fails energetic viability. Therefore energetic viability performs independent eliminative work and is not a corollary of consistency. □ □

Remark 28 (The Two-Teeth Consequence). This independence establishes that the Bedrock Constraint filters via two distinct mechanisms:

1. **Consistency Filter:** Eliminates forms that are structurally impossible (e.g., Liar Paradox, Square Circles).
2. **Solvency Filter:** Eliminates forms that are resource impossible (e.g., Infinite Archives, Laplace Demons, Unbounded Growth).

In physical domains, the solvency filter manifests as thermodynamic constraints (energy, entropy, matter). In formal domains, it manifests as bounded representability (finite specification, finite proof length, finite computational resources). Both are instances of the same structural requirement: forms cannot require unbounded resources relative to their substrate. Existence requires passing both.

6 Self-Application and Unavoidability

The constraint must apply to itself to be foundational. Any candidate foundation must remain determinate under self-application; otherwise it cannot function as a foundation.

Theorem 29 (Unavoidability by Stable Targeting)**.** *Any evaluable denial of the persistence constraint presupposes stable targeting of the denied claim across the denial act. Therefore no evaluable denial can succeed without satisfying the constraint at the level required for the denial to remain aimed at its target.*

Proof. Let K denote the persistence constraint claim. Consider an attempted denial $\neg K$.

For $\neg K$ to be evaluable as a denial of K, the denial act must preserve stable reference to K across the expression, interpretation, and assessment of the denial. That is, the denial must remain aimed at the same target throughout.

Stable targeting is a form of re-identification: there must exist a selective invariant that keeps the target fixed across admissible transformations involved in denial (tokening, paraphrase, inferential unpacking). By Definition 14, this requires re-identifiability.

But re-identifiability requires the persistence filter's consistency component (Theorem 25). Hence any evaluable denial of K presupposes the very consistency condition K asserts, at least at the level required to keep the denial aimed at K.

Therefore denial cannot succeed as a determinate denial without satisfying the constraint. □ □

6.1 The Performative Paradox

If persistence without contradiction is negotiable, show me a single determinate denial that does not presuppose stable reference and non-contradiction.

Anyone who tries to deny persistence without contradiction must already rely on it. If someone says "persistence without contradiction is not necessary," that statement must persist long enough to be understood, must refer to the same proposition from start to finish, and must not contradict itself while being evaluated. The denial therefore uses the very constraint it attempts to reject. For the denial to be about this claim, it must preserve stable targeting of the claim across the denial act.

If someone proposes an alternative foundation like measurement, consciousness, mathematics, or physics, the same test applies: can it be identified more than once as the same thing? If yes, it already presupposes persistence; if no, it cannot function as a foundation at all. This is not a physical law or a metaphysical opinion but a constraint on what can be coherently thought, said, or denied.

Remark 30 (Contrast with Self-Reference Failures). The Liar sentence ("This sentence is false") fails self-application precisely because it violates recursive closure: iteration produces oscillation, not stable identity. Our constraint, by contrast, maintains determinate content under iteration. This asymmetry is diagnostic: claims that fail self-application cannot serve as bedrock; claims that survive it are candidates for foundation.

7 Exhaustion of Alternatives

We now demonstrate that no more primitive foundation exists. Each alternative either explicitly presupposes persistence or collapses into indeterminacy without it.

Theorem 31 (No Foundational Regress)**.** *Any attempted foundation F that is evaluable as a foundation must be re-identifiable under its constitutive transformations, and therefore must*

satisfy the persistence filter. Hence persistence without GCC is not one foundation among competitors, but a precondition on any candidate foundation.

Proof. Let F be proposed as foundational. For F to function as a foundation, it must be a determinate target of reference across the act of grounding, comparison, or explanatory deployment. That requires F to be re-identifiable across the transformations constitutive of foundational use (inference, application, quotation, formalization). By Definition 14, this requires a selective stable criterion.

If F contains GCC at its identity-grounding layer, then by Lemma 24 F is not re-identifiable and thus cannot function as a determinate foundation.

Therefore any evaluable foundation must satisfy non-GCC persistence conditions. Since F was arbitrary, persistence without GCC is a precondition on all candidate foundations, eliminating foundational regress. □ □

7.1 Identity-First Approaches

Proposal: Perhaps identity is more primitive than persistence [6, 10, 4, 5].
Proposition 32 (Identity Presupposes Persistence)**.** *The identity relation cannot be formulated or evaluated without presupposing persistence.*

Proof. Consider an identity claim of the form $A = B$. To evaluate this claim, the following structural requirements must hold:

1. The term A must be referentially stable: we must be able to refer to A at the beginning of the evaluation and refer to the same entity at the end.
2. The term B must be referentially stable.
3. The relation symbol "=" must maintain stable meaning across the comparison.
4. The claim itself must persist as the same claim from assertion through evaluation to conclusion.

Each of these requires re-identification in the sense of Definition 14: a selective criterion that picks out the same entity across transformations (temporal progression, inferential steps, or evaluation stages). By Definition 10, re-identification presupposes persistence: the entity must satisfy $\psi \equiv_\psi (\hat{\delta} \circ \delta)^n(\psi)$ under its admissible transformations.

Therefore, to formulate or evaluate any identity claim $A = B$, we must presuppose that A, B, the relation "=", and the claim itself all persist. If persistence were derivative from identity, then identity claims could not be formulated without already presupposing their own derivative (persistence), creating either circularity or regress. Therefore identity cannot be more primitive than persistence. □ □

7.2 Distinction-First Approaches

Proposal: Perhaps bare distinction or difference is more primitive [2].

Proposition 33 (Distinction Presupposes Persistence)**.** *Distinction cannot obtain without presupposing persistence of the distinguished entities.*

Proof. Consider a distinction between entities X and Y, expressed as $X \neq Y$. For this distinction to be determinate, the following must hold:

1. There must exist a criterion C_X that picks out X and not Y.
2. There must exist a criterion C_Y that picks out Y and not X.
3. These criteria must remain stable across the act of distinguishing.

By Definition 14, criteria that pick out entities selectively and stably are precisely re-identification criteria. If X and Y do not persist, then no stable criteria C_X and C_Y can exist.

Therefore, distinction presupposes persistent boundaries, and cannot be more primitive than persistence. □ □

7.3 Iteration-First Approaches

Proposal: Perhaps iteration or recursion is more primitive.
Proposition 34 (Iteration Presupposes Persistence)**.** *Iteration cannot be formulated without presupposing persistence of the iterated operation and substrate.*

Proof. Consider an iteration expressed as $f^n(x)$ for some function f, initial value x, and iteration count $n \in \mathbb{N}$. For this iteration to be well-defined:

1. The function f must be the same function at each application.
2. The substrate x must persist as the same type of entity across applications.
3. The iteration count n must maintain stable meaning.
4. The composition operation itself must remain determinate.

Thus iteration presupposes re-identifiability of the operation and substrate across applications, hence persistence by Definition 14 and Definition 10. □ □

7.4 Temporal-First Approaches

Proposal: Perhaps time is more primitive, with persistence as mere temporal extension [9].
Proposition 35 (Time Presupposes Persistence)**.** *Temporal ordering cannot be formulated without presupposing persistence of events and temporal relations.*

Proof. A temporal ordering claim $e_1 < e_2$ requires determinate reference to e_1, e_2, and the ordering relation across the act of comparison. That requires re-identification criteria stable under the transformations constitutive of temporal comparison, hence persistence. □ □

7.5 Observer-First Approaches

Proposal: Perhaps observation or consciousness is more primitive [3].
Proposition 36 (Observation Presupposes Persistence)**.** *Observation cannot occur without presupposing persistence of both observer and observed.*

Proof. An observation act requires that both observer and observed remain determinately referable across the duration of observation and reporting, which entails re-identifiability and therefore persistence. □ □

7.6 Logic-First Approaches

Proposal: Perhaps logical laws are more primitive [7, 1].
Proposition 37 (Logic Presupposes Persistence)**.** *Logical inference cannot be formulated without presupposing persistence of propositions and inference rules.*

Proof. Inference requires that the same propositions and rules be re-identifiable across inferential steps; this is persistence as a structural condition. □ □

7.7 Information-First Approaches

Proposal: Perhaps information or probability is more primitive than persistence.
Proposition 38 (Information Presupposes Persistence)**.** *Information measures require stable distinguishability of states across application, and therefore presuppose persistence of the state space and its partition structure.*

Proof. A Shannon information measure requires a partition of possible states and stable assignment of probabilities to those states. Partition structure and probability assignment are only well-defined if the states remain re-identifiable under the transformations constitutive of measurement or inference. By Definition 14, this is persistence. Therefore information cannot be more primitive than persistence. □ □

Conclusion: By the foregoing propositions, every proposed alternative foundation presupposes persistence. No alternative is more primitive. The argument does not assume linear foundationalism. It establishes a constraint on any admissible foundational graph.

7.8 Exhaustion of Escape Routes: Non-Determinacy, Non-Discursive Rejection, and Definitional Collapse

A recurring objection is not that the proofs fail, but that the framework can be avoided by refusing one of its entry conditions. This section proves that the remaining escape routes are not counterexamples. They are either covertly determinate, or not truth-evaluable as rejections or existence claims.

7.8.1 Escape Route 1: "I Mean Existence Prior to Determinacy"

An objector may accept that persistence is required for determinate existence, but claim a more primitive notion of "raw" or "pre-conceptual" existence outside the scope of selective invariance.

We do not deny that one may gesture at such a notion. The claim here is eliminative and structural: if "existence" is asserted as a truth-evaluable claim about something, then it necessarily presupposes determinacy in the sense of stable selective targeting.

Theorem 39 (Non-Determinacy Cannot Be Asserted as Aboutness)**.** *Let C be a truth-evaluable claim of the form "X exists" (or "X exists non-determinately"). If C is about a target X (as opposed to being non-referential noise), then C presupposes a nontrivial selective criterion distinguishing X from at least one alternative and stable across the evaluation of C. Therefore, any existence claim that is about X presupposes determinate re-identifiability in the sense of Definition 14.*

Proof. Assume C is truth-evaluable and about X. For C to be about X, the term "X" must successfully target something rather than everything or nothing. Thus there must exist at least one alternative Y such that X is not interchangeable with Y relative to the targeting role of C; otherwise the claim has no determinate subject.

This yields a nontrivial selectivity requirement: there exists a criterion $C_X : S \to \{0, 1\}$ such that $C_X(X) = 1$ and $\exists Y$ with $C_X(Y) = 0$. Moreover, this targeting must remain stable across the transformations constitutive of utterance, interpretation, and evaluation of C; otherwise "X" fails to refer to the same target throughout the claim's evaluation. By Definition 14, this is precisely re-identifiability.

Therefore any truth-evaluable existence claim about X presupposes determinate re-identifiability of X under the relevant transformations. □ □

Remark 40 (Result of Theorem 39). The objection "I assert non-determinate existence" bifurcates: either it is a truth-evaluable claim about some target X, in which case it presupposes determinacy, or it is not truth-evaluable as about anything, in which case it is not a counterclaim within the space of evaluation.

7.8.2 Escape Route 2: "I Reject This Non-Discursively"

A second escape route attempts to avoid the self-application argument by refusing discursive participation (gesture, silence, lived experience). We grant that non-discursive acts occur. The question is whether such an act can function as a rejection of the persistence constraint.

Proposition 41 (Rejection Requires Determinate Targeting)**.** *Let K denote the persistence constraint claim. An act A counts as a rejection of K only if A is (i) intentionally directed at K, (ii) discriminable from non-rejection acts, and (iii) stable enough to be interpretable as carrying that directed content. If A satisfies (i)–(iii), then A presupposes stable selective targeting and therefore presupposes the persistence constraint at the level required for rejection to be determinate. If A does not satisfy (i)–(iii), then A is not a rejection of K.*

Proof. If A is to count as a rejection of K, it must be about K rather than about some unrelated object, which requires stable targeting of K across interpretation. It must also be distinguishable from acts that are not rejections, which requires nontrivial selection. Finally, it must remain stable enough to be interpretable as carrying that directed meaning. These conditions instantiate re-identification and selectivity in the sense of Definition 14 and Lemma 20. Therefore, if A is a determinate rejection, it presupposes the same persistence constraints required for any evaluable denial (Theorem 29). If A fails these conditions, it is not a rejection of K. □ □

7.8.3 Escape Route 3: "Your Definition Makes the Result Tautological"

A third escape route claims the framework is circular because Definition 1 builds selective invariance into the meaning of determinate existence.

The reply is structural: the definition is not an arbitrary stipulation, but the minimal operational criterion required for existence-talk to do any work at all. We formalize this as an operational collapse theorem.

Theorem 42 (Operational Collapse of Existence)**.** *Let $E(x)$ be any predicate intended to function as a truth-evaluable existence predicate within a discourse, meaning it supports:*

1. *nontrivial discrimination between existents and non-existents, and*
2. *stability of that discrimination under the role-constitutive transformations relevant to the discourse.*

Then E entails the existence of a nontrivial selective criterion stable under admissible transformations, i.e., the structure required by Definition 1 and Definition 14.

Proof. Assume E functions as an existence predicate in the stated operational sense.

If E fails nontrivial discrimination, then either $E(x)$ holds for all x (trivialism) or for no x (nihilism), and cannot perform the role of an existence predicate. Thus nontrivial selectivity is required: there exists X and Y such that $E(X) = 1$ and $E(Y) = 0$.

If E lacks stability under the role-constitutive transformations required to evaluate existence claims, then the truth-evaluable content of "X exists" cannot remain fixed across evaluation, since the targeted subject may drift under admissible transformations. This collapses existence-talk into indeterminacy and prevents E from serving as an operational existence predicate.

Therefore any operational existence predicate must support a selective criterion stable under admissible transformations, which is exactly the structure formalized by determinate re-identifiability and the persistence filter. □ □

7.8.4 Result: Escape-Proof Constraint for Truth-Evaluable Discourse

Corollary 43 (Rejection Dichotomy)**.** *Any attempted rejection of the persistence constraint is either:*

1. ***Evaluable***, *in which case it presupposes stable selective targeting and therefore presupposes the constraint (Theorem 29), or*
2. ***Non-evaluable***, *in which case it is not a rejection within the space of truth-evaluable claims.*

Proof. Immediate from Proposition 41 and Theorem 29. □ □

8 Re-Identification: Structural, Not Epistemic

8.1 The Core Distinction

A central confusion must be eliminated: re-identification is not observer recognition. It is structural invariance under transformation.

Re-identification is not about what we can know. It is about what structural conditions must obtain for a form to be the same across change.

8.2 Rejection of Epistemic Readings

This framework does not claim "We can only know what persists" (epistemology) or "Persistence requires observation" (idealism). It claims: persistence is a structural condition on forms, independent of minds, observers, or knowledge. A form that violates the constraint cannot persist even in a universe with no observers.

9 Addressing Standard Objections

9.1 Objection 1: Paraconsistent Logic

A common objection is that paraconsistent logics handle contradiction without explosion, potentially allowing for "persisting contradictions." We resolve this via structural diagnosis.

Theorem 44 (Paraconsistent Persistence Requires Stable Partition)**.** *Any paraconsistent formal system that persists as a determinate inferential practice avoids globally coupled contradiction at the level grounding its persistence by exhibiting a stable partition witness.*

Proof. Let $\mathcal{L}$ be a paraconsistent system that persists as a determinate inferential practice, meaning it maintains stable rule-identity and a nontrivial notion of validity across use.

Determinacy of rule-identity and validity requires selectivity: there must exist a criterion that distinguishes valid from invalid inferences and preserves that distinction across admissible rule applications. By Lemma 20, such selectivity induces a nontrivial partition at the grounding level of persistence.

If contradictions were globally coupled at that grounding level, then by Lemma 23 no stable re-identification criterion could exist relative to the admissible transformations constitutive

of the system's identity. That would destroy determinacy of rule-identity and validity, contradicting the assumption that $\mathcal{L}$ persists as a determinate inferential practice.

Therefore any persisting paraconsistent system must exhibit some stable partition witness at the grounding level, even if its object language represents contradictions. □ □

9.2 Objection 2: Quantum Superposition

Statement: Quantum mechanics allows particles to be in superpositions like $|\uparrow\rangle + |\downarrow\rangle$. This seems to violate the constraint.

Structural Diagnosis: Superposition is not a classical contradiction; treating it as such is a category error. This paper's claim is about contradictions at the identity-grounding layer of a form relative to its role transformations. Quantum mechanics illustrates the constraint: superposition states are perfectly determinate quantum states satisfying all persistence requirements. What QM reveals is that classical predicate structure (simultaneous sharp position and momentum) generates globally coupled contradictions via the uncertainty principle. Nature enforces the constraint: such states don't exist. Only quantum states satisfying the persistence filter exist.

9.3 Objection 3: Instantaneous Existence

Statement: Perhaps something can exist instantaneously without persisting.

Structural Diagnosis: Even instantaneous reference presupposes minimal persistence. To identify "a particle at time t" requires that the concept "particle" and the location t persist across the act of identification. The claim "X exists at t" must persist long enough to be asserted. Instantaneous existence is not outside persistence but minimal persistence.

9.4 Falsifiability: The Persistence Trilemma

The constraint is falsifiable through the following exhaustive case analysis. Any attempted counterexample to the persistence constraint must fall into exactly one of three categories:

1. **The contradiction is partitionable.** There exists a partition witness π (via indexing, staging, localization, or hierarchical confinement) that preserves a selective invariant. Result: Not a counterexample.

2. **The contradiction is unpartitionable.** No admissible partition witness exists relative to $\Delta(\psi)$. By Lemma 24, no selective invariant can exist. Result: The form does not persist.

3. **The form persists without selectivity.** The object is claimed to persist while lacking any selective invariant. Result: The claim collapses into non-discriminating noise.

10 Paradox Resolution

10.1 Worked Examples with Formal Analyses

This section applies the definitions in Sections 2–4 as a reproducible diagnostic workflow. Each case proceeds by: (i) fixing role R, (ii) fixing role-constitutive transformations $\Delta(\psi)$, (iii) identifying the contradictory predicates $P, \neg P$, and (iv) testing whether a stable partition witness π exists under $\Delta(\psi)$ (Definition 16).

10.2 The Liar Paradox (Full Worked Example)

Let ψ be the sentence token L with content: "This sentence is false."

Role: truth-bearer. Thus $\Delta(\psi)$ includes evaluation operations (Proposition 8).

Predicates: Let $P(\psi)$ be "ψ is true" and $\neg P(\psi)$ be "ψ is false." By the content of L, evaluation yields:

$$P(L) \Rightarrow \neg P(L), \qquad \neg P(L) \Rightarrow P(L).$$

Thus the evaluation transformation forces both $P(L)$ and $\neg P(L)$ at the grounding layer for role identity as a truth-bearer.

Partition test: Any proposed partition witness π must separate P and $\neg P$ by one of Localization, Indexing, Staging, or Confinement while remaining stable under the evaluation operations in $\Delta(\psi)$.

- Localization fails: the same sentence token is the locus of the predicate.
- Indexing fails under role-constitutive evaluation: the contradiction reappears at each evaluation index.
- Staging fails: the evaluation stage that produces $P(L)$ immediately forces $\neg P(L)$ and conversely, so no stage-stable separation exists.
- Confinement fails: there is no consistent superstructure grounding a determinate truth-bearer role while isolating the contradiction to a proper subdomain.

Therefore no stable π exists under $\Delta(\psi)$, so the contradiction is globally coupled (Definition 16). By Theorem 25, L cannot persist as a determinate proposition.

10.3 Russell's Paradox (Full Worked Example)

Let ψ be the set $R = \{x \mid x \notin x\}$.

Role: set. Thus $\Delta(\psi)$ includes membership tests (Proposition 8), in particular the transformation $\delta_\in$ that evaluates whether $\psi \in \psi$.

Predicates: Let $P(\psi)$ be "$\psi \in \psi$" and $\neg P(\psi)$ be "$\psi \notin \psi$." By definition of R:

$$R \in R \iff R \notin R.$$

Thus membership testing, a role-constitutive operation, forces $P(R)$ and $\neg P(R)$ as inseparable requirements at the identity-grounding layer for ψ as a set.

Partition test: A stable partition witness π would need to separate P and $\neg P$ under the membership-test transformations in $\Delta(\psi)$.

- Localization fails: the membership query is about the same object R.
- Indexing fails: any index that permits the membership test reproduces the biconditional.
- Staging fails: the membership test itself is the constitutive stage; it cannot be postponed while preserving the role.
- Confinement fails: any attempted hierarchy that isolates the contradiction changes the role (it produces a different object than the naive comprehension set R).

Hence no stable π exists under $\Delta(\psi)$, so R contains GCC. By Theorem 25, R does not exist as a determinate set.

10.4 The Sorites Paradox (Worked Example)

Let ψ be the predicate "heap" applied across grain-removal iterations.

Role: classifier predicate over physical aggregates.

Transformations: $\Delta(\psi)$ includes small perturbations (remove one grain), and evaluation of membership in the predicate extension under those perturbations.

Diagnosis: The paradox arises because no stable selective criterion C_{heap} is fixed under the role-constitutive perturbations. The predicate is used as if it denotes a single sharp boundary, but its admissible perturbations generate boundary drift. The failure is not GCC in an underlying object; it is failure of determinate re-identifiability for the claimed boundary. Thus the paradox indicates that "heap" is not a single determinate form without an explicit stabilizing criterion.

10.5 Worked Examples: GCC vs. Partitionable Contradiction (Section 10.5)

This subsection provides a fully uniform diagnostic suite to demonstrate that the framework is not post-hoc labeling but executable structure.

10.5.1 Uniform Diagnostic Template

Each case follows the same six steps:

1. **Candidate form** ψ.
2. **Role** R.
3. **Role-constitutive transformations** $\Delta(\psi)$ (Proposition 8).

4. **Grounding predicates** $P(\psi)$ and $\neg P(\psi)$ (or boundary predicate when non-GCC).
5. **Partition test** (Localization, Indexing, Staging, Confinement) for a stable witness π (Definition 17).
6. **Verdict** (GCC eliminated, partitioned survives, or non-GCC failure mode).

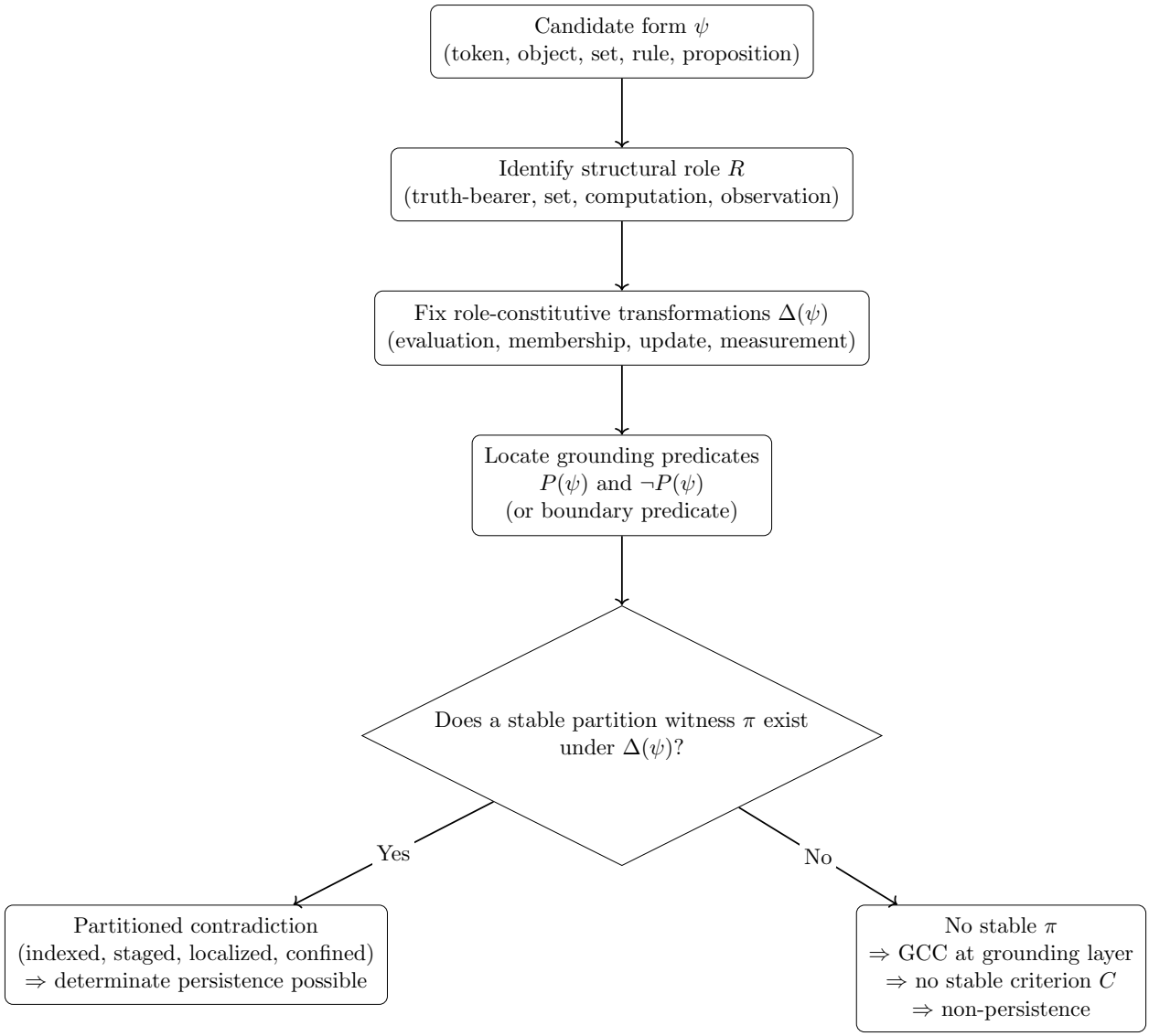

Figure 1: Diagnostic pipeline for contradiction cases: determine whether contradiction is globally coupled (GCC) or partitionable under role-constitutive transformations.

10.5.2 Worked Example 1: Liar (GCC)

1) Candidate form ψ: sentence token L: "This sentence is false."

2) Role R: truth-bearer.

3) $\Delta(\psi)$: evaluation operations (truth-assignment).

4) Predicates: $P(L)$: "L is true"; $\neg P(L)$: "L is false." Evaluation forces

$$P(L) \Rightarrow \neg P(L), \qquad \neg P(L) \Rightarrow P(L),$$

so $P(L)$ and $\neg P(L)$ are coupled at the grounding layer for its truth-bearer role.

5) Partition test:

- Localization fails: one token carries both conditions.
- Indexing fails: re-evaluation at any index reproduces the coupling.
- Staging fails: each evaluation stage forces its negation.
- Confinement fails: isolating the contradiction removes truth-bearer determinacy.

6) Verdict: No stable π. GCC. Eliminated by Theorem 25.

10.5.3 Worked Example 2: Russell (GCC)

1) Candidate form ψ: $R = \{x \mid x \notin x\}$.

2) Role R: set.

3) $\Delta(\psi)$: membership tests, including $\delta_\in$ for $R \in R$.

4) Predicates: $P(R)$: "$R \in R$"; $\neg P(R)$: "$R \notin R$." By definition,

$$R \in R \iff R \notin R,$$

so membership testing forces both.

5) Partition test:

- Localization fails: same object queried.
- Indexing fails: any admissible membership-test index reproduces the biconditional.
- Staging fails: membership testing is constitutive of set-role.
- Confinement fails: hierarchy changes the role from naive set to a different object.

6) Verdict: No stable π. GCC. Eliminated by Theorem 25.

10.5.4 Worked Example 3: Sorites (Boundary Drift, Non-GCC)

1) Candidate form ψ: predicate "heap" over aggregates.

2) Role R: classifier predicate on physical collections.

3) $\Delta(\psi)$: small perturbations (remove one grain) plus membership evaluation under those perturbations.

4) Predicates: not a single $P(\psi) \wedge \neg P(\psi)$ at one grounding layer. The failure is: the boundary predicate cannot remain stable under admissible perturbations without an explicit stabilizer.

5) Partition test: not applicable as GCC, because the issue is not an inseparable P and $\neg P$ requirement but underdetermination of a stable criterion.

6) Verdict: Not GCC. Indeterminate predicate unless a stabilizing criterion is fixed (threshold, context parameter, or supervaluation family).

10.5.5 Worked Example 4: Barber (GCC)

1) Candidate form ψ: barber B defined by: "B shaves all and only those men who do not shave themselves."

2) Role R: classifier plus actor definition.

3) $\Delta(\psi)$: role-constitutive application of the defining condition to candidate instances, including self-application to B.

4) Predicates: $P(B)$: "B shaves B"; $\neg P(B)$: "B does not shave B." Definition forces

$$P(B) \iff \neg P(B).$$

5) Partition test:

- Localization fails: single agent.
- Indexing fails: time-indexed shaving preserves the rule coupling under definition application.
- Staging fails: definition evaluation is constitutive of being this barber.
- Confinement fails: isolating the contradiction changes the role claim.

6) Verdict: No stable π. GCC. Eliminated by Theorem 25.

10.5.6 Worked Example 5: Berry (GCC unless Confined)

1) Candidate form ψ: phrase "the smallest positive integer not definable in under eleven words."

2) Role R: definability predicate over $\mathbb{N}$ inside one language.

3) $\Delta(\psi)$: word-counting, definability checking within the same language, and minimum-selection.

4) Predicates: Let $P(n)$ be "n is definable in under eleven words." Define

$$b := \min\{n \in \mathbb{N} \mid \neg P(n)\}.$$

But the phrase itself supplies a definition (in many renderings) implying $P(b)$ while selection requires $\neg P(b)$. Thus $P(b) \wedge \neg P(b)$ at the grounding layer of the naive definability role.

5) Partition test:

- Localization fails: non-spatial.
- Indexing fails: mere parameter indexing does not remove self-definability in the same language.
- Staging fails: the defining act is the constitutive stage.
- Confinement succeeds only if one introduces an explicit metalanguage boundary (object vs meta definability), which changes the role claim from "flat definability" to "stratified definability."

6) Verdict: GCC for naive flat definability. Eliminated by Theorem 25. Fixed only by confinement (language stratification), which is a role restriction.

10.5.7 Worked Example 6: Grelling–Nelson (GCC)

1) Candidate form ψ: predicate H meaning "does not describe itself" (heterological).

2) Role R: predicate over predicates.

3) $\Delta(\psi)$: role-constitutive self-application.

4) Predicates: Define $H(x) \iff \neg x(x)$. Then

$$H(H) \iff \neg H(H),$$

forcing $P(H)$ and $\neg P(H)$ at the grounding layer for its self-application role.

5) Partition test:

- Localization fails: not spatial.
- Indexing fails: any admissible index preserves the biconditional.
- Staging fails: self-application is constitutive.
- Confinement fails unless one changes the role to forbid self-application, which is not the same predicate role.

6) Verdict: No stable π. GCC. Eliminated by Theorem 25.

10.5.8 Worked Example 7: Curry (Selection Collapse / Trivialism)

1) Candidate form ψ: sentence C: "If this sentence is true, then Q" where Q is arbitrary.

2) Role R: truth-bearer in a system with conditional introduction and modus ponens.

3) $\Delta(\psi)$: evaluation plus inference rules (including conditional reasoning) as role-constitutive transformations for truth-bearer use inside the inferential practice.

4) Predicates: Let $P(C)$ be "C is true"; content yields $P(C) \Rightarrow Q$. Standard reasoning yields derivability of Q from C under assumptions that preserve $P(C)$ as a truth status for inference.

5) Partition test:

- Localization fails: one token.
- Indexing fails: the inferential coupling is not removed by time or context indices while keeping the same inferential role.
- Staging fails: the derivation stage is constitutive of using C as a truth-bearer in the system.
- Confinement succeeds only by restricting the conditional or truth predicate so that the derivation does not go through, which changes the role-constitutive inference package.

6) Verdict: The characteristic failure is *selection collapse*: everything becomes derivable (trivialism) at the criterion level, which violates selectivity (Definition 14, clause 2). In the present framework, that is elimination as determinate truth-bearer content unless the role is restricted (e.g., typed truth predicates or restricted contraction), which is confinement by changing admissible transformations.

10.5.9 Worked Example 8: Ship of Theseus (Criterion Underdetermination, Non-GCC)

1) Candidate form ψ: a persisting object under part-replacement (a ship).

2) Role R: macroscopic physical object with identity across time.

3) $\Delta(\psi)$: role-constitutive transformations include gradual replacement of parts, reassembly operations, and tracking across time.

4) Predicates: the paradox does not force $P(\psi) \wedge \neg P(\psi)$ at a single grounding layer. Rather, multiple admissible identity criteria compete (same matter, same structure, same continuity, same function) without a fixed selection rule inside the role description.

5) Partition test: not a GCC partition problem. The failure mode is that the identity condition $\equiv_\psi$ is underdetermined relative to the admissible transformations.

6) Verdict: Not GCC. It is criterion underdetermination. Persistence is recoverable by selecting and fixing an identity criterion (which amounts to specifying the intended role more precisely).

10.5.10 Worked Example 9: Unexpected Hanging (Index/Stage Conflation, Non-GCC)

1) Candidate form ψ: announcement of an "unexpected" execution within a known window.

2) Role R: epistemic constraint statement governing agents' permissible knowledge states.

3) $\Delta(\psi)$: iterative reasoning updates across days, plus public announcement uptake, plus knowledge-closure steps assumed by the paradox.

4) Predicates: the apparent contradiction arises from conflating: (i) object-level scheduling facts, and (ii) meta-level knowledge about what can be deduced at each stage. It is not a single object that must satisfy $P(\psi)$ and $\neg P(\psi)$ at one grounding layer.

5) Partition test: the standard resolution is indexing or staging: separate knowledge-at-day-d from knowledge-at-day-$(d+1)$ and separate public announcement content from agents' derived meta-claims. The paradox relies on collapsing these indices and stages.

6) Verdict: Not a clean GCC in an object. It is index/stage conflation. Persistence is restored by explicit indexing of epistemic states and by restricting closure assumptions across stages.

10.5.11 Worked Example 10: Grandfather (GCC under Single-History Constraints)

1) Candidate form ψ: a single-history timeline H plus an intervention action I (time travel) that targets an ancestor.

2) Role R: single-history causal model with global consistency constraints (one lineage per history).

3) $\Delta(\psi)$: causal evolution plus the intervention operator $do(I)$ applied within that same single-history model.

4) Predicates: Let $P(H)$ be "the agent exists in H" and let $Q(H)$ be "the grandfather exists long enough to generate the lineage that produces the agent." The intervention is defined to enforce $\neg Q(H)$. But agent existence presupposes $Q(H)$ under single-history lineage constraints. Thus the model requires $P(H)$ and $\neg Q(H)$ while $P(H) \Rightarrow Q(H)$ at the causal-grounding layer.

5) Partition test:

- Localization fails: not spatial.
- Indexing fails if all indices refer to one single consistent history.
- Staging fails if causal grounding is global across stages (lineage is a cross-stage constraint).
- Confinement succeeds only by changing the role: branching histories (many-worlds), fixed-point consistency constraints (Novikov), or prohibiting the intervention. Those are different admissible transformation families.

6) Verdict: Under single-history global consistency, no stable π exists. The contradiction is GCC at the causal-grounding layer. Eliminated unless the role is changed (branching or fixed-point confinement).

10.5.12 Worked Example 11: Venus Contrast (Partitioned by Indexing)

1) Candidate form ψ: Venus as one astronomical object with two observational reports.

2) Role R: astronomical object under observation.

3) $\Delta(\psi)$: observations at different indices plus cross-identification operations.

4) Predicates: Let $P(\psi, i)$ be "bright at index i" and $\neg P(\psi, i)$ be "not bright at index i". Consider indices i_m (morning) and i_e (evening) such that $P(\psi, i_m)$ and $\neg P(\psi, i_e)$.

5) Partition test: Indexing provides a stable witness:

$$\pi := \{i_m, i_e\}, \qquad P(\psi, i_m), \qquad \neg P(\psi, i_e).$$

π is stable under observational transformations and preserves identity of Venus across the index family.

6) Verdict: Partitioned contradiction (Indexing). Survives. Not GCC.

10.5.13 Diagnostic Summary

Table 1 consolidates the diagnostic outcomes.

Case	**Failure Mode**	**Verdict**	**Partition Witness / Fix**
Liar (L)	GCC	Eliminated	None exists
Russell (R)	GCC	Eliminated	None exists
Sorites (heap)	Boundary drift	Indeterminate	Requires stabilizer criterion
Barber (B)	GCC	Eliminated	None exists
Berry (b)	GCC (flat def.)	Eliminated/Fixed	Confinement via metalanguage
Grelling–Nelson	GCC	Eliminated	None exists
Curry (C)	Selection collapse	Eliminated/Fixed	Confinement via role restriction
Ship of Theseus	Underdetermination	Indeterminate	Fix $\equiv_\psi$ explicitly
Unexpected Hanging	Index/stage conflation	Resolved	Explicit staging/indexing
Grandfather	GCC (causal)	Eliminated/Fixed	Confinement via branching
Venus	Partitioned	Survives	Indexing (i_m, i_e)

Table 1: Uniform diagnostics applied to classic paradoxes and contrast cases.

11 Scope and Non-Parochial Power

11.1 Mathematical Ontology

Corollary 45 (Mathematical Determinacy is Intensional)**.** *Mathematical determinacy is intensional: objects are accessible as finitely specifiable invariants under admissible transformations.*

Consequently:

- *The Peano Axioms exist (finite rule specification).*
- *The natural numbers $\mathbb{N}$ exist as the pattern generated by successor iteration (finite rule).*
- *"A completed, actualized infinite list of all natural numbers" does not exist as a determinate form in any finite representational substrate; infinities exist as intensional generators.*

Proof. Energetic viability requires $\Phi_\psi(n) \geq W_\psi(\delta, n)$. In abstract domains, W_ψ is the representational cost to maintain the structure's boundary. A finite rule has bounded maintenance cost. A completed infinite extension has unbounded maintenance cost. Therefore finite rules persist; actualized infinities do not as determinate forms in any finite medium. □ □

11.2 Computation

Computational systems demonstrate the constraint directly. A program state persists if it transitions determinately without self-contradiction. The energetic viability condition appears directly: computation requires physical resources (energy, memory, time).

12 Falsification Conditions

The constraint would be refuted by demonstration of any of the following:

1. **Persisting GCC:** A form that persists as determinate while containing globally coupled contradiction at the grounding level with no partition witness stable under its constitutive transformations.
2. **Re-identification Without Selectivity:** A form that is re-identifiable across transformations while lacking any selective criterion.
3. **Solvency-Free Persistence:** A form with unbounded maintenance cost that persists indefinitely on finite sustaining capacity.
4. **Self-Exception:** Demonstration that this constraint itself contains GCC or violates energetic viability at the level of its own statement.

13 CRIS as Necessary Consequence

Proposition 46 (CRIS Structure)**.** *Any persisting form necessarily exhibits:*

- ***Consistency****: no globally coupled contradiction at the grounding level,*
- ***Recursion****: stability under self-application of its identity-defining boundary,*
- ***Invariance****: maintenance of identity across admissible transformations,*

- ***Selection****: nontrivial boundaries that exclude incompatible states.*

Proof. **Consistency:** By Theorem 25, globally coupled contradiction prevents persistence.

Recursion: Persistence requires boundary stability under re-application (Definition 10).

Invariance: Re-identification (Definition 14) requires invariance of a criterion under admissible transformations.

Selection: Re-identification requires a selective criterion; by Lemma 20, selectivity implies a nontrivial partition separating the form from alternatives.

Therefore any persisting form exhibits CRIS structure. □ □

14 Conclusion

The foundation of reality is not a particle, force, field, axiom, or process. It is a constraint:

Only what admits stable selective re-identification under its role-constitutive transformations can exist as a determinate form.

This is not a hypothesis. It is the structural precondition for determinate existence. Any entity claiming to exist must satisfy this constraint or fail to be a determinate entity at all. Globally coupled contradiction destroys the partition structure required for selectivity, therefore it cannot ground existence. The constraint is thus ontological: it separates determinate being from non-being by eliminative closure.

References

[1] Beall, J.C., & Restall, G. (2009). *Logical Pluralism*. Oxford University Press.

[2] Deleuze, G. (1994). *Difference and Repetition*. Columbia University Press.

[3] Kant, I. (1781/1998). *Critique of Pure Reason*. Cambridge University Press.

[4] Kripke, S. (1980). *Naming and Necessity*. Harvard University Press.

[5] Lewis, D. (1986). *On the Plurality of Worlds*. Blackwell.

[6] Lowe, E.J. (2006). *The Four-Category Ontology: A Metaphysical Foundation for Natural Science*. Oxford University Press.

[7] Priest, G. (2006). *In Contradiction: A Study of the Transconsistent* (2nd ed.). Oxford University Press.

[8] Tarski, A. (1944). The Semantic Conception of Truth: and the Foundations of Semantics. *Philosophy and Phenomenological Research*, 4(3), 341–376.

[9] Whitehead, A.N. (1929). *Process and Reality*. Macmillan.

[10] Wiggins, D. (2001). *Sameness and Substance Renewed*. Cambridge University Press.